Shear stiffening associated with angular forces

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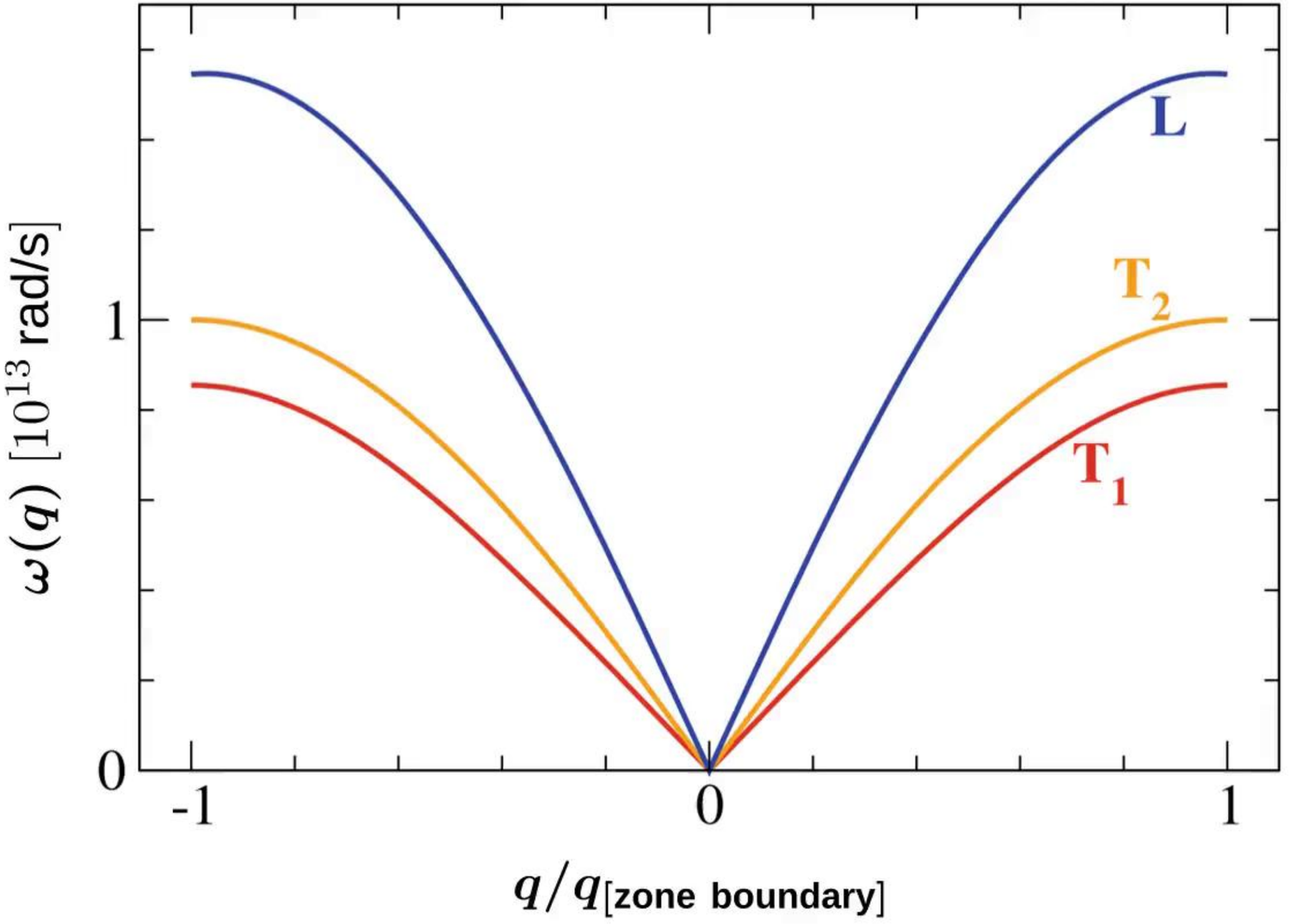
Typical phonon dispersion curves

Long wave-length approximation:

$$\omega(q) \lesssim 10^{12}\text{-}10^{13} \text{ rad/s}.$$

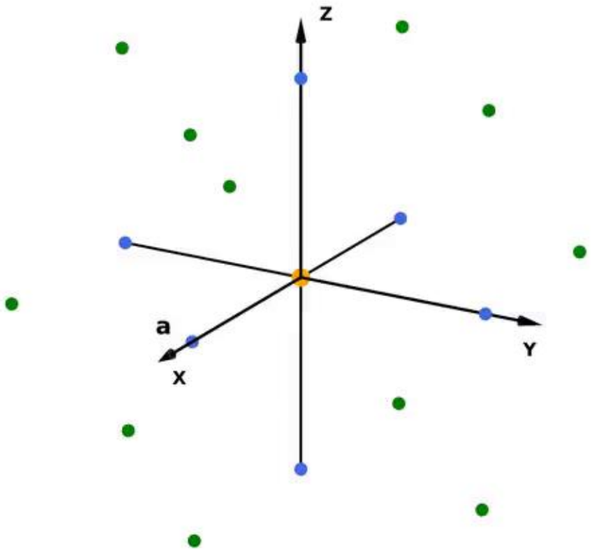
- $\omega(q) = v \cdot q$ for $q \to 0$.

- Wave velocities from the elastic constants.

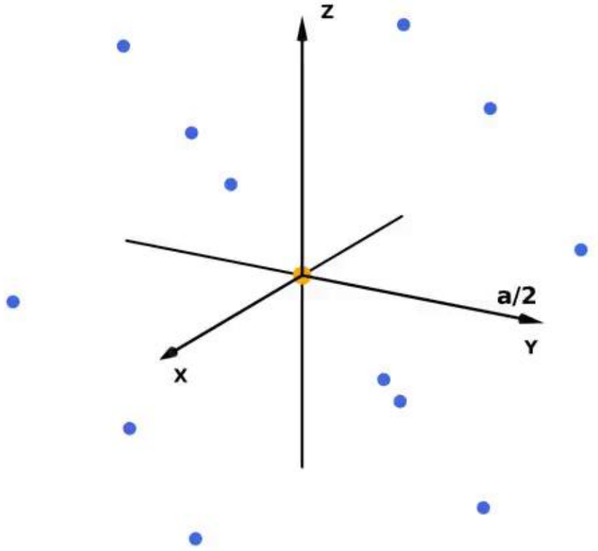


Crystal model

Simple Cubic (SC):



Face-Centered Cubic (FCC):



$$U_{TOT} = U_{LJ} + U_{TB}.$$

Potential

U$_{LJ}$:

$$U_{LJ}(r_{ij}) = \sum_{\text{pairs } ij} V_{LJ}(r_{ij}) = \sum_{i=1}^{N} \left[\frac{1}{2} \sum_{j<i} V_{LJ}(r_{ij}) \right]$$

$$V_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^{6} \right].$$

γ_w:

$45° \leftrightarrow w = 1.$ $\quad 120° \leftrightarrow w = 4$.

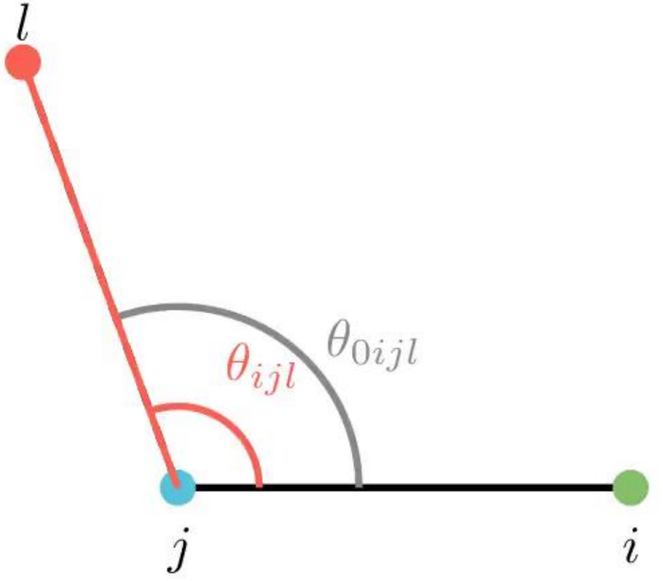
$60° \leftrightarrow w = 2.$ $\quad 135° \leftrightarrow w = 5$.

$90° \leftrightarrow w = 3.$ $\quad 180° \leftrightarrow w = 6$.

U$_{TB}$:

$$U_{TB} = \sum_{\substack{i,j,l \\ i<j<l}} \gamma(\theta_{0ijl}) \left[1 - \cos(\theta_{ijl} - \theta_{0ijl}) \right]$$

$$\equiv \sum_{\substack{i,j,l \\ i<j<l}} \gamma_w \left[1 - \cos(\theta_{ijl} - \theta_{0ijl}) \right].$$



Energy density

ϵ :

$$x' = (1+\epsilon_{xx})\hat{x} + \epsilon_{xy}\hat{y} + \epsilon_{xz}\hat{z}.$$
$$y' = \epsilon_{yx}\hat{x} + (1+\epsilon_{yy})\hat{y} + \epsilon_{yz}\hat{z}.$$
$$z' = \epsilon_{zx}\hat{x} + \epsilon_{zy}\hat{y} + (1+\epsilon_{zz})\hat{z}.$$



$\epsilon \to e$

$$u \equiv u - u_0 = \frac{U}{V} = \frac{1}{2}\tilde{C}_{\lambda\mu}e_\lambda e_\mu.$$

$r \to r'$:

$$R = r' - r = \xi(r)\,\hat{x} + \psi(r)\,\hat{y} + \zeta(r)\,\hat{z}.$$

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{44} \end{bmatrix}.$$

$$= \frac{1}{2}C_{11}(e_{xx}^2 + e_{yy}^2 + e_{zz}^2) + \frac{1}{2}C_{44}(e_{yz}^2 + e_{zx}^2 + e_{xy}^2) + C_{12}(e_{yy}e_{zz} + e_{zz}e_{xx} + e_{xx}e_{yy}).$$

Elastic constants

$-\boldsymbol{e_{xx} \neq 0}$

$\rightarrow u(e_{xx}) = \frac{1}{2} C_{11}\, e_{xx}^2.$

$-\boldsymbol{e_{xx} = e_{yy} = e \neq 0}$

$\rightarrow u(e) = (C_{11} + C_{12})\, e^2.$

$-\boldsymbol{e_{xy} \neq 0}$

$\rightarrow u(e_{xy}) = \frac{1}{2} C_{44}\, e_{xy}^2.$

SC:

$$\begin{cases} C_{11}^{SC} = \frac{1}{r_0^3}\left[72\epsilon + 4(\gamma_1 + \gamma_2 + 2\gamma_3 + \gamma_4 + \gamma_5)\right]. \\ C_{12}^{SC} = -\frac{2}{r_0^3}(\gamma_1 + \gamma_2 + 2\gamma_3 + \gamma_4 + \gamma_5). \\ C_{44}^{SC} = \frac{2}{r_0^3}\left(\gamma_1 + 2\gamma_2 + 6\gamma_3 + 2\gamma_4 + \gamma_5\right). \end{cases}$$

FCC:

$$\begin{cases} C_{11}^{FCC} = \frac{\sqrt{2}}{r_0^3}\left[72\epsilon + 4(\gamma_2 + 2\gamma_3 + \gamma_4)\right]. \\ C_{12}^{FCC} = \frac{\sqrt{2}}{r_0^3}\left[36\epsilon - 2(\gamma_2 + 2\gamma_3 + \gamma_4)\right]. \\ C_{44}^{FCC} = \frac{4\sqrt{2}}{r_0^3}\left(9\epsilon + \gamma_2 + \gamma_4\right). \end{cases}$$

Sound velocities

$$\rho\, v^2 \begin{bmatrix} \xi_0 \\ \psi_0 \\ \zeta_0 \end{bmatrix} = \Gamma(\vec{n}) \begin{bmatrix} \xi_0 \\ \psi_0 \\ \zeta_0 \end{bmatrix} .$$

$$\Gamma(\vec{n}) = \begin{bmatrix} C_{11}n_x^2 + C_{44}(n_y^2 + n_z^2) & (C_{12} + C_{44})n_x n_y & (C_{12} + C_{44})n_x n_z \\ (C_{12} + C_{44})n_x n_y & C_{11}n_y^2 + C_{44}(n_x^2 + n_z^2) & (C_{12} + C_{44})n_y n_z \\ (C_{12} + C_{44})n_x n_z & (C_{12} + C_{44})n_y n_z & C_{11}n_z^2 + C_{44}(n_x^2 + n_y^2) \end{bmatrix} .$$

Special directions

SC:

$$[1\,0\,0] \quad T > L : \gamma_3 > 18\,\epsilon + \frac{\gamma_1 + \gamma_5}{2}$$

$$[1\,1\,0] \quad T_1 > L : \gamma_3 < -18\,\epsilon - \frac{\gamma_1 + \gamma_2 + \gamma_4 + \gamma_5}{2}$$

$$T_2 > L : \gamma_3 < -\frac{\gamma_2 + \gamma_4}{4}$$

$$[1\,1\,1] \quad T > L : \gamma_3 < -\frac{\gamma_2 + \gamma_4}{4}.$$

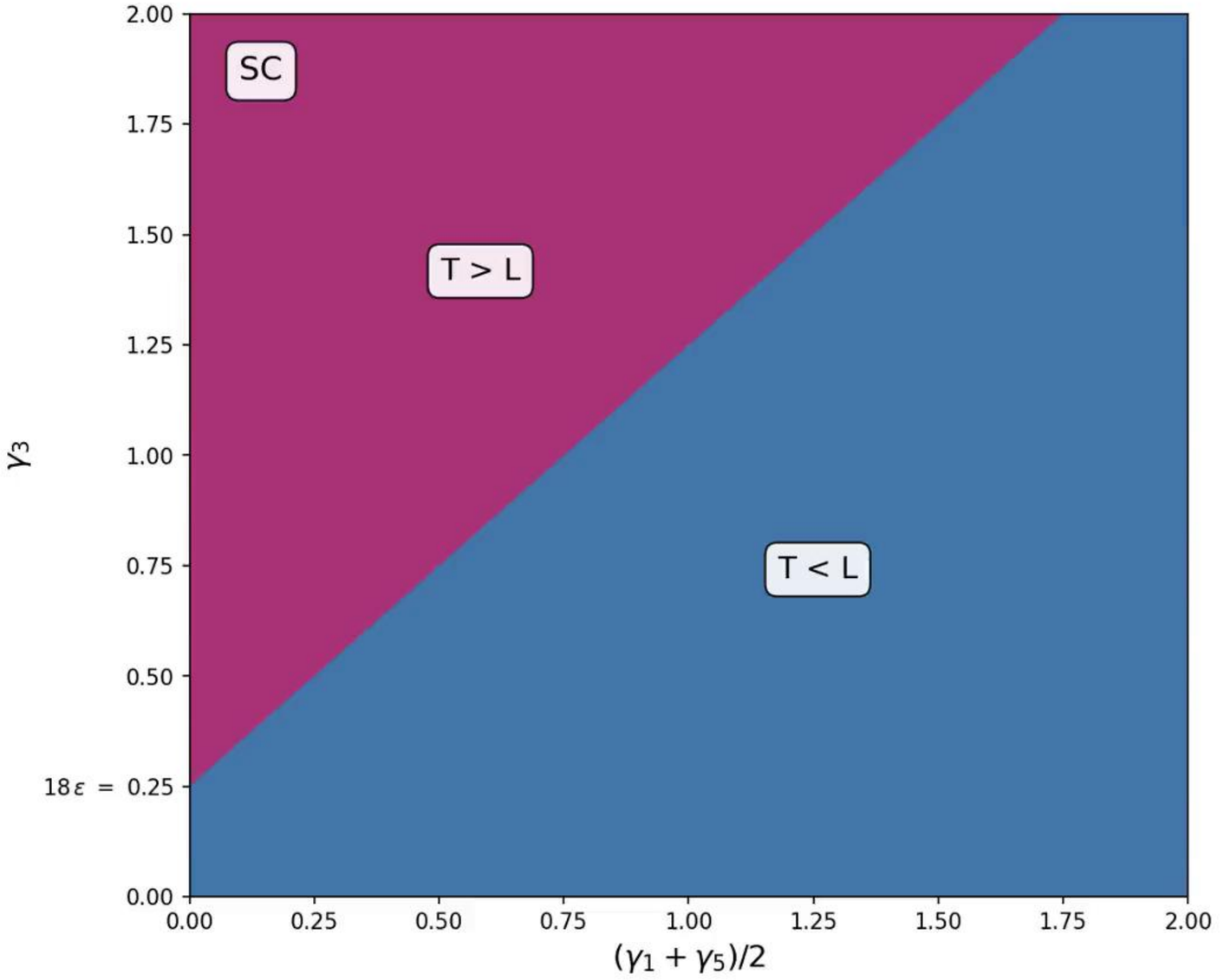
FCC:

$$[1\,0\,0] \quad T > L : \gamma_3 < -\frac{9}{2}\,\epsilon$$

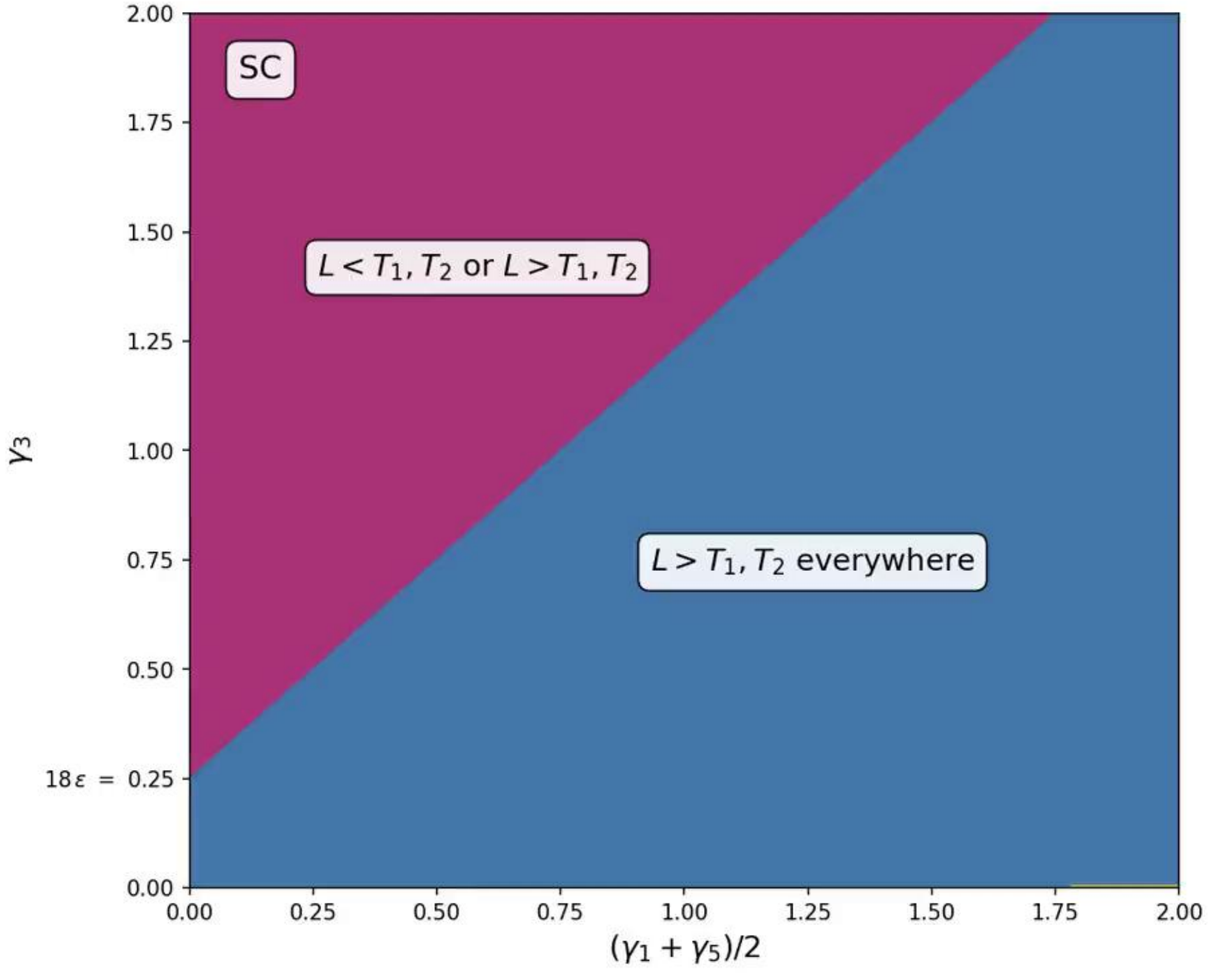
$$[1\,1\,0] \quad T_1 > L : \gamma_3 < -27\,\epsilon - \frac{\gamma_2 + \gamma_4}{2}$$

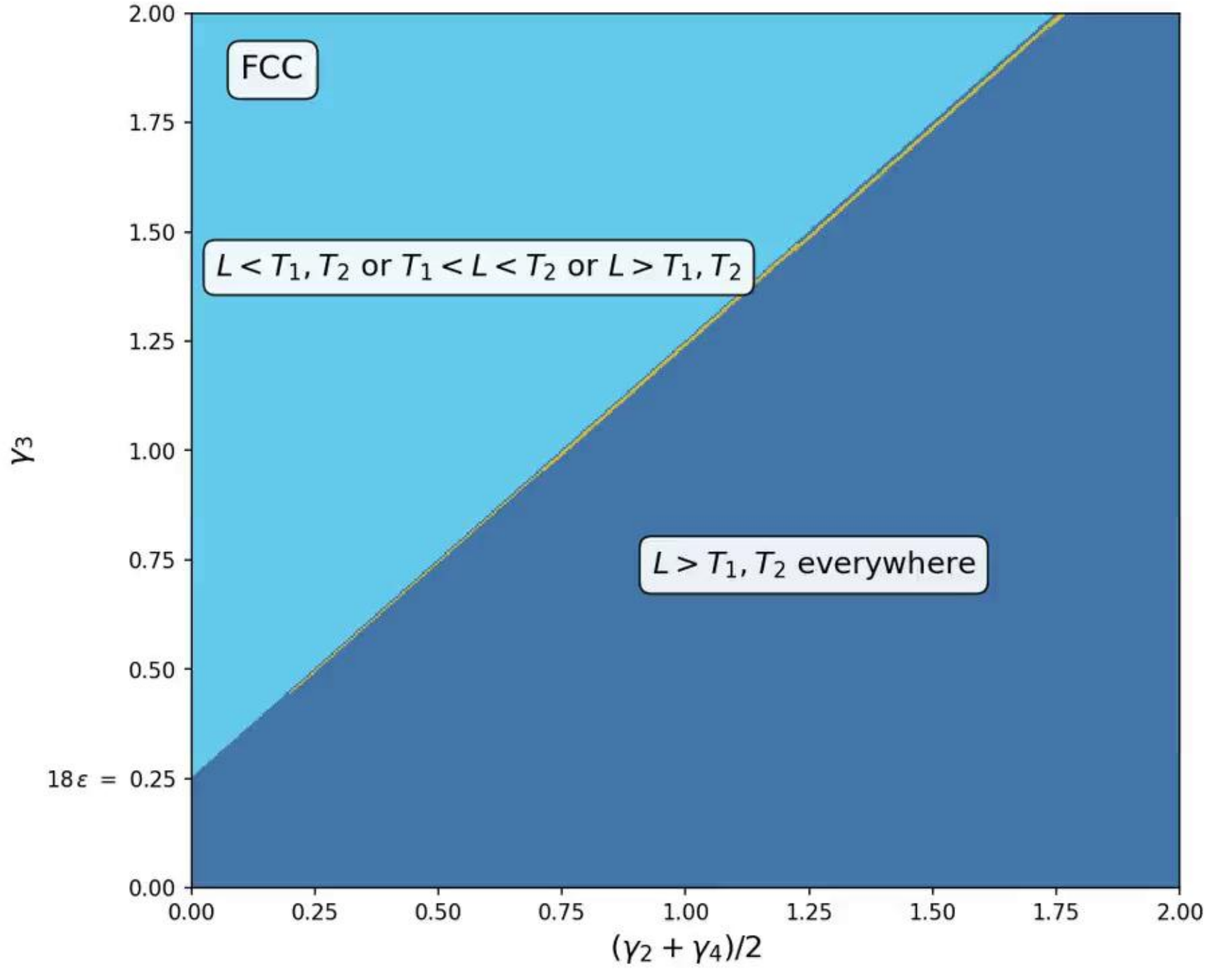
$$T_2 > L : \gamma_3 > 18\,\epsilon + \frac{\gamma_2 + \gamma_4}{2}$$

$$[1\,1\,1] \quad T > L : \gamma_3 > 18\,\epsilon + \frac{\gamma_2 + \gamma_4}{2}\,.$$



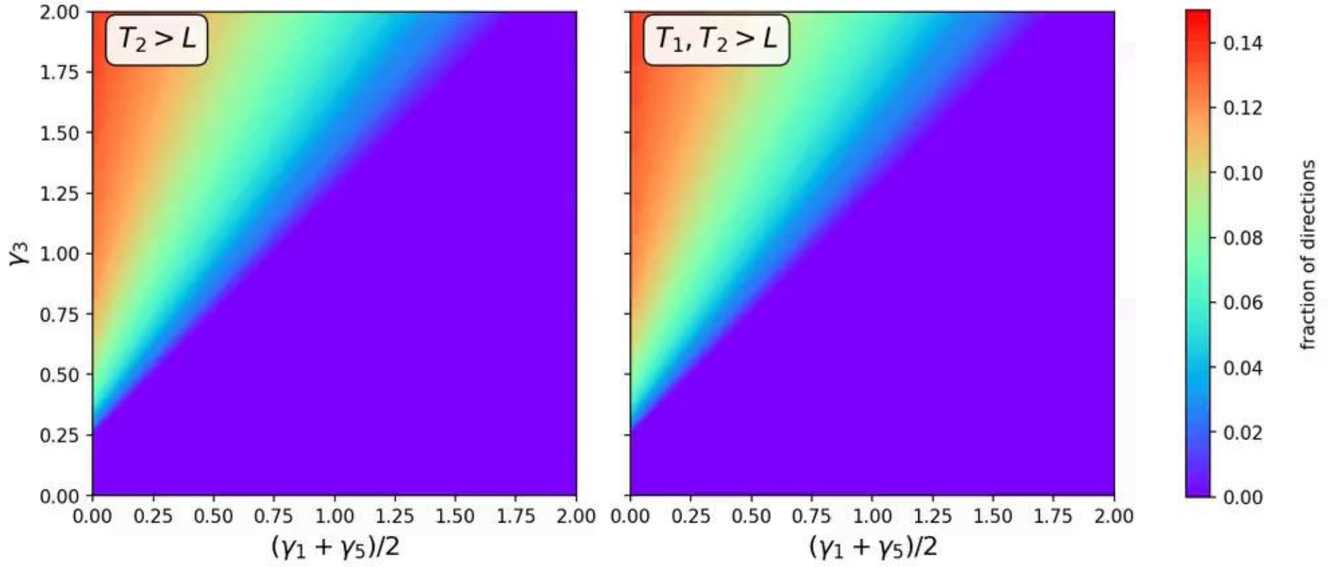
General $\vec{q}$ phase diagrams



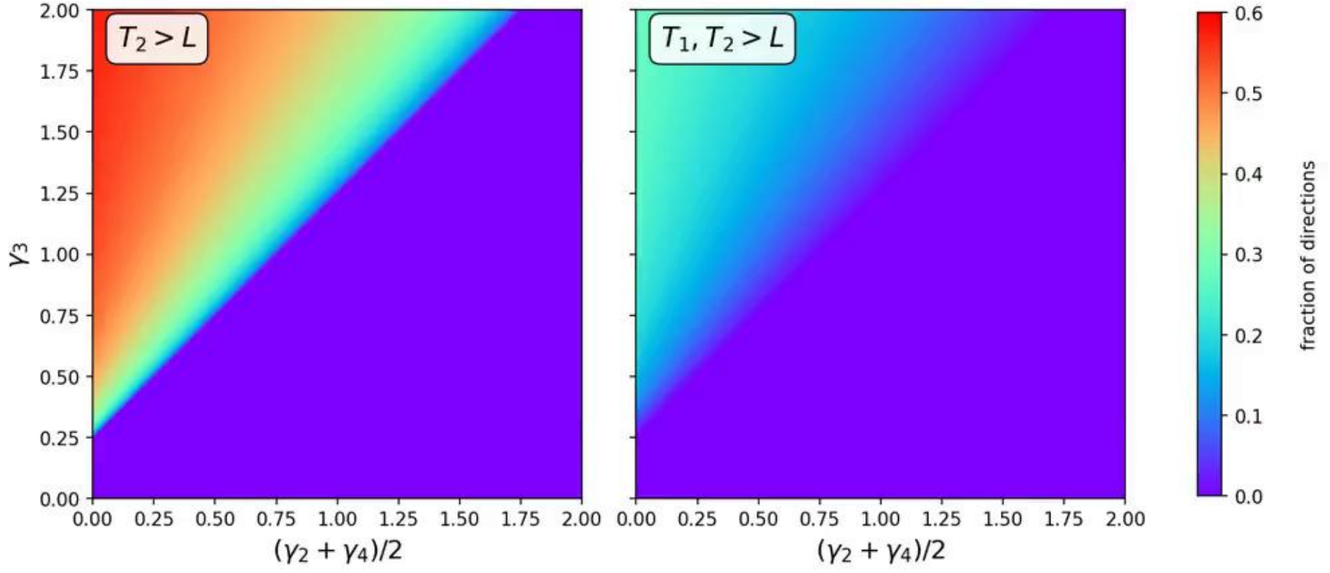


Fraction of directions

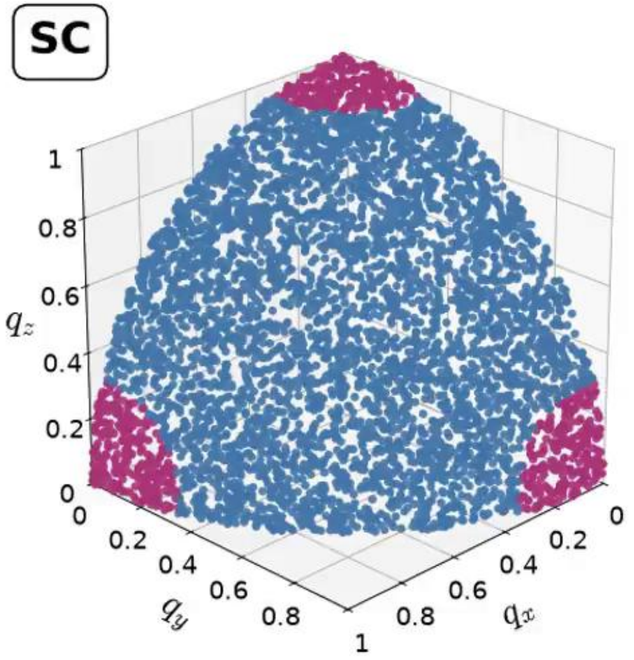
SC:

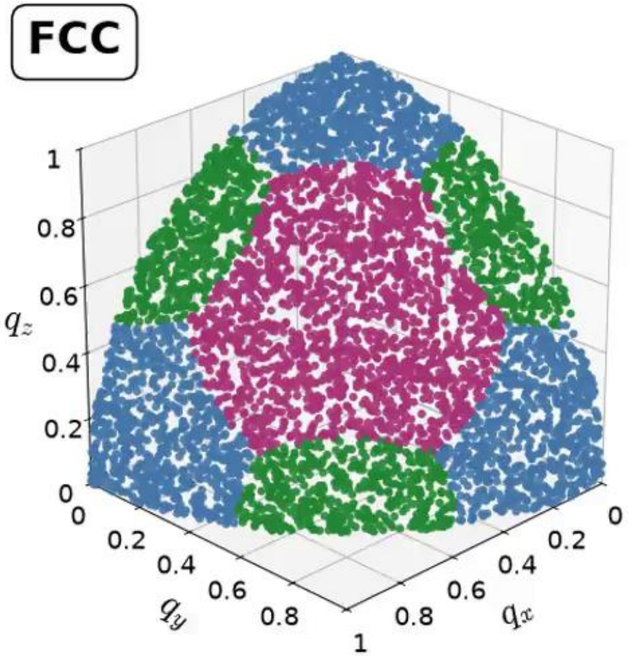


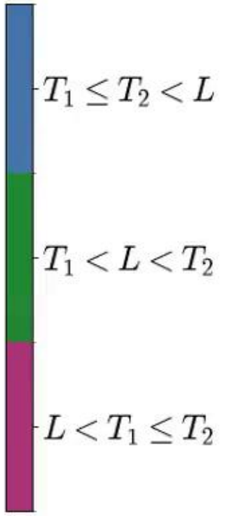
FCC:



Random sampling of the unit sphere







$$\gamma_1 = \gamma_5 = 1.$$
$$\gamma_3 = 50.$$

$$\gamma_2 = \gamma_4 = 1.$$
$$\gamma_3 = 50.$$

Conclusions

- In the adopted model, regions of the parameter space where the velocity inversion occurs have been found

- No region was found where $v_L < v_{T_1}, v_{T_2}$ for every propagation direction simultaneously

Future developments

- Extend the analysis to other lattices, starting from BCC
- Add second-neighbor angular terms for FCC, as done for SC
- Enlarge the parameter space: independent parameters

for each non-equivalent angular term

- Replace fixed neighbor lists with a flexible force field

(distances and angles on the fly, Stillinger-Weber style)